1. Describe one result you found interesting doing your task.

One thing I found interesting was that the two project options use machine learning in different ways even though both focus on turbulent transport. Group 1A aims to learn the relationship between TGLF inputs and predicted fluxes so that a surrogate can produce results in milliseconds. Group 1B focuses on improving TGLF's saturation rule using CGYRO data. This helped me see how our earlier interest in HPC and machine learning could become a more specific project. I was especially interested in comparing regression, trees, and neural networks to see how well they capture the same input-output relationship.

2. Describe one thing you struggled with doing your task.

The biggest difficulty was understanding the difference between approximating TGLF's predictions and improving the underlying saturation rule. Both options involve learning from plasma data, so I had to think carefully about what each model would actually be trained to predict. I also struggled with judging which direction would be manageable while we are still learning the physics. The slides' discussion of correlated inputs and sparse data made it clear that choosing a model is only part of the problem; we also need to understand what the available data can support.

3. Identify one choice that you made in your investigation. Why did you make this choice?

One choice I made was to join Group 1A. I chose this direction because building a surrogate for TGLF fits my interest in machine learning and gives us a clear starting point: exploring the inputs and fluxes before comparing predictive models. I liked that we could begin with simpler regression or tree models and then evaluate whether a neural network offers an improvement. Group 1B was another reasonable option, but I felt more interested in learning the full mapping from plasma parameters to fluxes. This also gives our group a concrete way to connect data exploration with our eventual modeling work.